WORKED SOLUTION

STEP III 2010, Question 10

Mechanics • SHM

STEP III 2010, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/10-s3-q10

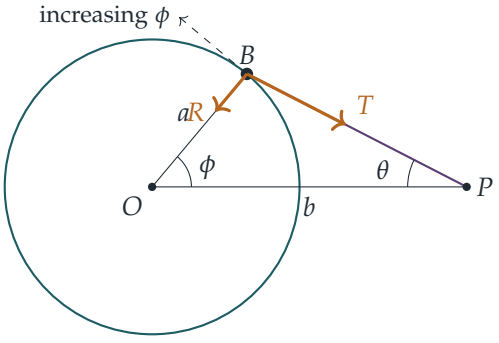
The key idea

The bead is confined to a circle, so only the component of force along the tangent to the ring changes its motion; the normal reaction (and gravity, which the ring's vertical reaction balances) never enters. Everything else is the geometry of the triangle OBP, and the sine rule turns that geometry into the string length and, for small angles, into the linear relation between θ and ϕ that the second part asks for.

Solution

The equation of motion

Model the bead as a particle on a smooth ring and the string as light and elastic. Take P on the positive x-axis and measure ϕ anticlockwise, so that B is at angle ϕ from OP, as in the figure. In triangle OBP the angle at O is ϕ, the angle at P is θ, and so the angle at B is $\pi - \theta - \phi$.



Let $l = BP$. By the sine rule in triangle OBP,

$$\frac{l}{\sin\phi} = \frac{a}{\sin\theta} = \frac{b}{\sin(\theta+\phi)},$$

using $\sin(\pi-\theta-\phi)=\sin(\theta+\phi)$. In particular $l=\dfrac{a\sin\phi}{\sin\theta}$.

The string is taut throughout, because the closest B can get to P is $b-a$, and $b-a>c$. So by Hooke's law the tension is

$$T=\frac{\lambda(l-c)}{c}=\lambda\left(\frac{a\sin\phi}{c\sin\theta}-1\right).$$

The forces in the horizontal plane are the tension T along BP and the normal reaction R of the smooth ring along the radius OB. (Vertically, the weight is balanced by the vertical part of the reaction.) The bead moves on a circle of radius a, so its acceleration along the tangent, in the direction of increasing ϕ, is $a\ddot{\phi}$. Only T has a tangential component.

The string BP makes angle $\pi-\theta-\phi$ with BO, which is the inward radius. So it makes angle $\theta+\phi$ with the outward radius, and its component along the tangent has size $T\sin(\theta+\phi)$. That component points towards P, which is the direction of *decreasing* ϕ. Resolving along the tangent,

$$ma\ddot{\phi}=-\lambda\left(\frac{a\sin\phi}{c\sin\theta}-1\right)\sin(\theta+\phi),$$

as required. (For $\phi<0$ the figure is reflected in OP; then $\theta<0$ too, and the same equation holds with signed angles.)

Small oscillations

The sine rule gave $a\sin(\theta+\phi)=b\sin\theta$ exactly. When θ and ϕ are small, $\sin x\approx x$ on both sides, so

$$a(\theta+\phi)\approx b\theta.$$

Hence $\theta\approx\dfrac{a\phi}{b-a}$ and $\theta+\phi\approx\dfrac{b\theta}{a}\approx\dfrac{b\phi}{b-a}$.

> **Ask yourself** What happens to $\dfrac{a\sin\phi}{c\sin\theta}$ as both angles go to zero? It looks like $0/0$.

That quotient is just l/c, and as B approaches the equilibrium position the length l tends to $b-a$. Equivalently, $\dfrac{a\sin\phi}{\sin\theta}\approx\dfrac{a\phi}{\theta}\approx b-a$ from the relation above. So to leading order the tension is the constant equilibrium tension

$$T_0=\lambda\left(\frac{b-a}{c}-1\right)=\frac{\lambda(b-a-c)}{c},$$

which is positive since $c<b-a$. The factor $\sin(\theta+\phi)$ is already small, so any correction to T only contributes at second order.

Substituting into the equation of motion and keeping first-order terms,

$$ma\ddot{\phi}\approx-\frac{\lambda(b-a-c)}{c}\cdot\frac{b\phi}{b-a},\qquad\text{so}\qquad\ddot{\phi}=-\omega^2\phi,\quad\omega^2=\frac{\lambda b(b-a-c)}{mac(b-a)}.$$

This is simple harmonic motion, and the period of small oscillations about $\theta = \phi = 0$ is

$$\frac{2\pi}{\omega} = 2\pi\sqrt{\frac{mac(b-a)}{\lambda b(b-a-c)}}.$$

Checks

- Dimensions: λ is a force, so $mac(b-a)/(\lambda b(b-a-c))$ has dimensions $\mathrm{M\,L}/(\mathrm{M\,L\,T}^{-2}) = \mathrm{T}^2$, as a period should.
- As $c \to b-a$ the equilibrium tension vanishes and the period becomes infinite: a slack string gives no restoring force, which makes sense.
- Numerically, with $a = 1$, $b = 2.5$, $c = 0.7$, $\lambda = 3$, $m = 0.5$, the formula gives a period of 1.8586. Stepping the exact motion (with the force computed directly from the position of B) from a small displacement gives the same value to six significant figures. The exact equation of motion also agrees with a direct vector resolution at random positions.

Common slips

- Getting the sign wrong: the tangential pull is towards P, that is towards $\phi = 0$, so it must appear with a minus sign.
- Taking the tension to be zero at equilibrium. The string is stretched to length $b-a > c$ there, so the tension is $\lambda(b-a-c)/c$, and it is this tension that supplies the restoring force.
- Using $\theta + \phi \approx \phi$. Here θ is of the same order as ϕ, so dropping it gives the wrong period.

Beyond STEP

The method here is the standard way to find the frequency of small oscillations: find an equilibrium, linearise the equation of motion about it, and read off ω^2. A neat feature is that the restoring force comes entirely from the *equilibrium* tension acting through the change of direction of the string, which is the same mechanism as a stretched guitar string. In a first-year course this becomes Lagrangian mechanics and normal modes; Kleppner and Kolenkow, *An Introduction to Mechanics*, is a good place to see small-oscillation analysis developed carefully.

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